

Group Assignment - Micromouse

MTRN3100 - UNSW School of Mechanical and Manufacturing Engineering

Changelog

- v1.1 - 12/06/25 Updated the specification for Barebones movement task.
- v1.2 - 20/06/25 Fixed the maze indexing example in the maze representation section

Learning Outcomes

- Apply relevant theoretical knowledge about mobile robots, including locomotion, perception, and localisation, utilising onboard sensors, navigation, and path planning, for complex problem-solving.
- Apply general computer vision techniques for feature/object detection and tracking, for complex problem-solving.
- Demonstrate hands-on skills in mechatronics design, fabrication, and implementation by completing practical activities.
- Collaborate effectively within a team via participation in a problem-solving competition.

1 Objective

The objective of the micromouse challenge is to build a robot that navigates through a maze in the shortest time. Each team's run will be recorded and ranked to decide an overall winner.

1.1 Robot Specification

- The robot must be powered by the provided battery pack, with the appropriate fuse.
- All software processing must be done with the onboard Arduino nano microcontroller unless otherwise specified (computer vision for path generation).
- The robot must fit within a cylinder 150 mm wide and 150 mm tall.
- The robot must not use permanent fastening techniques e.g. epoxy and superglue.

Sufficient robot components are supplied however, you are allowed to include your own components within reason. If you are unsure if additional components are allowed, please contact the course staff. Replacement parts will not be provided unless you can prove your component was damaged through no fault of your own.

1.2 Maze representation

You are free to represent the maze however you choose. However, it must be autonomously generated from your computer vision solution. A source and destination location will be provided. The complete maze comprises cells in a 10 x 10 configuration. Each cell is 180mm x 180mm and the grid is enclosed (Figure. 1).

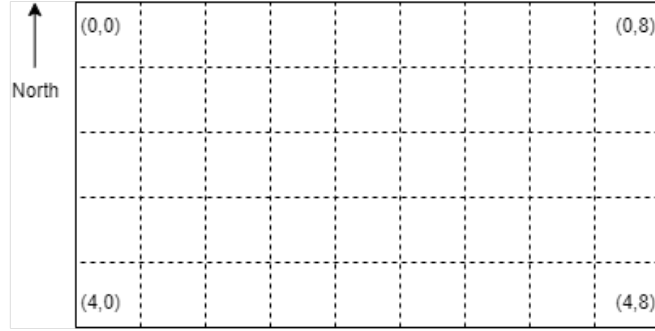


Figure 1: Representation of the maze grid with coordinates.

The start position is given in the configuration (row, column, direction), which indicates the robot's initial row and column (zero-indexed) as well as direction (N, S, E, W). If the start position is (1,6,N) the robot will be starting in the 2nd row, 7th column, pointing north. The goal position is given in the format (row, column), which indicates the target row and column.

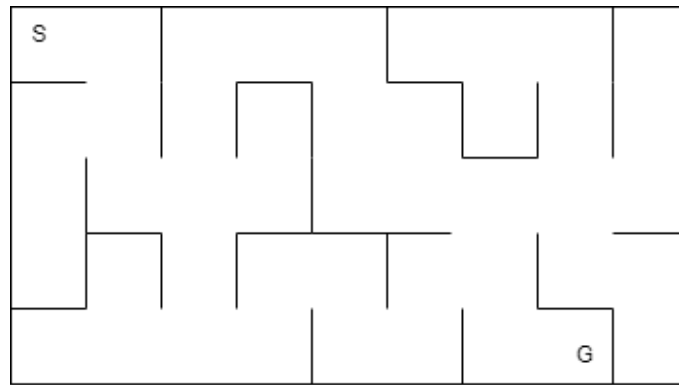


Figure 2: Example maze layout for the final competition. S represents the start location at cell (0,0). G represents the goal location at cell (4,7).

2 Barebones movement (Week 4) - 3% Total

The first milestone involves showing autonomous straight movement on any platform. A complete robot is not required; you simply need two actuated wheels on any movable platform to achieve locomotion. You must be able to demonstrate your ability to drive a distance of 200mm within a reasonable amount of time. This assessment will be marked during your week 4 lab time, and is worth 3% of your final course mark.

A 1m wide start line will be taped on the ground with two 1m wide lines taped 125mm and 275mm respectively from the start line, which define the goal zone. You will place your robot along the start line such that both wheels are touching the tape. You must then turn your robot on and have it drive to the goal zone before coming to rest. If all parts of your robot are within the goal zone when your robot comes to rest, you achieve 3 marks. If any part of your robot is touching or above the goal lines when your robot comes to rest, you achieve 1.5 marks. The testing time will be capped at one minute; exceeding this will result in 0 marks for the run. Manual intervention at any point after starting the robot will result in 0 marks for the run. Each team receives a maximum of 2 runs for this task, with your best attempt used for marking.

3 Simple driving (Week 8) - 7% Total

The simple driving task involves 3 different tasks. You may compile and upload code between trials, however, no significant changes will be allowed. A significant change is one requiring more than 5 minutes to implement. This assessment will be marked during your week 8 lab time, and is worth 7% of your final course mark.

3.1 Driving - 3%

In this task, you will need to use the front lidar and a distance-based controller to accurately move to a fixed distance from a wall. This task consists of three sequential challenges that must be completed in the same program (no

restarting between challenges). 1 mark will be awarded for each completion of each challenge. If your robot requires manual intervention at any stage or does not reach the desired state within 10 seconds, the run will be terminated.

First, the robot will be placed 200mm away from a static wall, It must drive forward until it is 100mm $\pm$ 5mm from the wall and at rest. Second, the demonstrator will slowly slide the wall back 100mm, and the robot must follow it, trying to maintain a distance of 100mm before eventually coming to rest 100mm $\pm$ 5mm from the wall. Finally, the demonstrator will slowly slide the wall forward 100mm, and the robot must follow it, trying to maintain a distance of 100mm before eventually coming to rest 100mm $\pm$ 5mm from the wall.

The demonstrator will not deliberately move the wall into the robot, however, you must achieve the desired setpoint within 10 seconds after the wall is moved into place. Each team receives a maximum of 2 runs for this task, with your best attempt used for marking.

3.2 Turning - 3%

In this task, you must be able to demonstrate your ability to accurately turn, either with lidars and/or an IMU. This test takes place within a single cell with walls on all 4 sides. This task consists of three sequential challenges that must be completed in the same program (no restarting between challenges). 1 mark will be awarded for each completion of each challenge. If your robot requires manual intervention at any stage or does not reach the desired state within 10 seconds, the run will be terminated.

First, your robot must perform a 90-degree clockwise turn when you turn it on. You will receive 1 mark if the robot is still and within $\pm$ 3 degrees of the desired rotation. Secondly, the demonstrator will raise the robot 50mm above the ground and turn it counter-clockwise up to 120 degrees. The demonstrator will lower the robot onto the floor, where the robot must rotate back to the setpoint. You will receive 1 mark if the robot is still and within $\pm$ 3 degrees of the desired rotation. Finally, the demonstrator will raise the robot 50mm above the ground and turn it clockwise up to 120 degrees. The demonstrator will lower the robot onto the floor, where the robot must rotate back to the setpoint. You will receive 1 mark if the robot is still and within $\pm$ 3 degrees of the desired rotation.

Each team receives a maximum of 2 runs for this task, with your best attempt used for marking.

3.3 Chaining Movements - 1%

You will be provided a string of 8 movements that your robot must be able to execute in a standard maze environment. For the maze shown in Figure. 2, the robot may start at (1,1,S) and be provided the string "lfrfflfr". This string would be the command sequence "left, forward, right, forward, forward, left, forward, right". In which forward indicates moving one cell, Left is a turn 90 degrees counter-clockwise, and Right is a turn 90 degrees clockwise.

Achieving at least 4 actions is worth 0.5 marks, while all 8 actions are worth the full 1 mark. A collision will terminate your run. Each team receives a maximum of 2 runs for this task, with your best attempt used for marking.

3.4 Submission

Marking will occur during your Week 8 Lab. Teams will be called to the front to be marked. It is expected that you turn up to the lab ready to get marked. Marks will be awarded at the discretion of the demonstrator. At a minimum, one team member must be present. If no team members are available, please contact the course staff. A team evaluation task will be available on Moodle after the completion of this task.

All the code used for the simple driving task must be pushed to your team's GitHub repository in a branch called "simple driving".

4 Micromouse (Week 12) - 20% Total

The micromouse task involves 3 different tasks. You may compile and upload code between trials, however, no significant changes will be allowed. A significant change is one requiring more than 5 minutes to implement. Figure. 2 shows an example of what the maze may look like for the simple driving assessment. This assessment will be marked during your week 12 lab time, and is worth 20% of your final course mark.

4.1 Micromouse race - 10%

This task involves path planning through computer vision and the autonomous maze solving competition.

4.1.1 Path Generation - 2%

Teams will take an image of the maze and use it to generate a series of commands that solve the maze. This series of commands can be copied to the robot's source code and uploaded. Teams may choose to use the command format laid out previously or a custom solution. The initial and goal locations will be provided to you in the format outlined in 1.2.

2 marks will be awarded for showing a demonstrator that your code executes. You must show the image input and the output to verify that your solution is not hard-coded. You will not be permitted to tune the values after taking an image, you must immediately execute the code and enter the output into the robot's source code in front of your demonstrator.

The generated path from this task is required for the next task, as such, if your program fails to execute or output a path, you will not be able to attempt 4.1.2.

4.1.2 Maze Completion - 8%

The path generated from 4.1.1 is entered in the robot's source code and uploaded to the robot as mentioned. The robot is then placed at the starting position and must begin navigating the maze when turned on. Marks are awarded based on maze completion and the time taken to solve the entire maze. The time is calculated from the moment the robot moves autonomously from the start to the moment it is fully enclosed in the final cell (facing the correct direction) and stationary.

Maze Completion %	Mark
[0%-20%)	0
[20%-40%)	1
[40%-60%)	2
[60%-80%]	3
[80%-99%]	4
[100%]	5

Table 1: Marks awarded for maze completion.

Speed	Mark
Top 50% of teams which completed the maze	1
Top 25% of teams which completed the maze	2
Top 10% of teams which completed the maze	3

Table 2: Marks awarded for competition speed.

To be eligible for the speed marks you must complete the maze. Marks for maze completion progress will be at the discretion of the marker. Each team will get a total of two runs. The first minor collision (hit but continues without assistance) will be allowed. The second minor collision or the first major collision (requires human intervention) will cause the run to terminate.

4.2 Continous Planning - 5%

For this task, a 5*5 cell section of the maze will be replaced by an obstacle course that the robot must navigate (Figure. 3). All the obstacles will be cylindrical with a diameter of 100mm at the same height as the maze walls, however, they will be placed randomly within the designated area. Computer vision should be used to generate an occupancy map from the image and solve the path. You may hardcode the position of cells in which the obstacle course is. The exact formatting used to transfer commands from the computer vision code to the robot is up to you. It may be best to generate a list of waypoints relative to each other, which the robot can move to in series. There will only ever be one entrance and exit to the obstacle course.

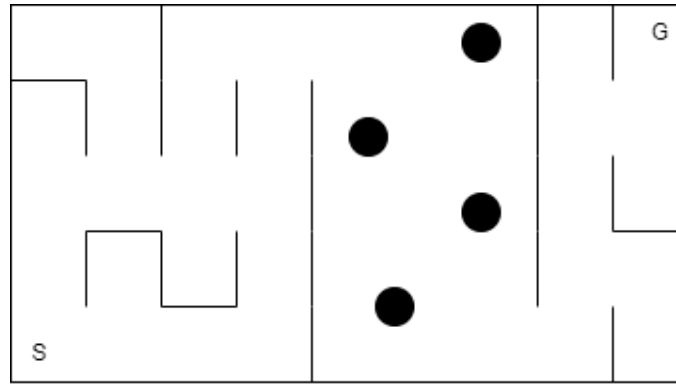


Figure 3: Example of a continuous maze your team may be presented with

1 mark will be awarded for generating an occupancy map and solving it. To receive this mark you must show an image with the trajectory to be taken by the robot to the demonstrator. 2 marks will be awarded for showing your robot can complete the obstacle course by following the trajectory shown in your occupancy map image. For this, your robot may start at the first cell in the obstacle course and finish in the last cell of the obstacle course. 2 marks will be awarded if the robot starts from the start of the maze and makes it to the end of the maze, moving through the obstacle course autonomously.

Each team will get a total of two runs. The first minor collision (hit but continues without assistance) will be allowed. The second minor collision or the first major collision (requires human intervention) will cause the run to terminate. Partial marks will be awarded at the discretion of the marker based on progress through the obstacle course if the run fails.

4.3 Autonomous Mapping - 5%

For this task, teams will use the robot to autonomously search and generate a map of the standard maze. It will then navigate to the start position, solve the maze, and complete the shortest path run of the maze. During the mapping of the maze, a visualisation of the map must be shown on the screen with a % completion score, derived from the number of cells the robot has visited.

The only information allowed to be hard-coded into your solution is the start coordinates (including orientation) and the goal coordinates. You must prove to your demonstrator that your solution is not hard-coded to get marks for this task. Marks for maze completion progress will be at the discretion of the marker. Each team will get a total of two runs. The first minor collision (hit but continues without assistance) will be allowed. The second minor collision or the first major collision (requires human intervention) will cause the run to terminate.

Maze Completion %	Mark
[0%-20%)	0
[20%-40%)	1
[40%-60%)	2
[60%-80%)	3
[80%-100%)	4

Table 3: Maze Completion Marks

1 mark will then be awarded if the robot solves the maze in the shortest path after mapping it. It is possible to explore and solve the shortest path maze without fully mapping the maze. In this case, teams will be awarded the full 5 marks.

NOTE: Attempting and succeeding in this task does not award marks for the computer vision path planning task.

4.4 Submission

Marking will occur during your Week 12 Lab. Teams will be called to the front to be marked. It is expected that you turn up to the lab ready to get marked. You may compile and upload code between trials, however, no significant changes will be allowed. A significant change is one requiring more than 5 minutes to implement. At a minimum, one team member must be present. If no team members are available, please contact the course staff. A team evaluation task will be available on Moodle after the completion of this task.

Code used for the micromouse Challenge must be pushed to your team's GitHub repository in a branch called "Micromouse".

5 Plagiarism

If you are unclear about the definition of plagiarism, please refer to [What is Plagiarism? — UNSW Current Students](#). You could get zero marks for the assignment if you were found:

- Knowingly providing your work to anyone and it was subsequently submitted (by anyone), or
- Copying or submitting any other person's work, including code from previous students of this course (except general public open-source libraries/code). Please cite the source if you refer to open-source code.

You will be notified and allowed to justify your case before such a penalty is applied.

5.1 Generative AI

Code written with a generative AI (ChatGPT) is allowed but must be clearly labelled with in-line comments and mentioned in the file header.